Curriculum Vitae

Yilang Xu

School of Earth and Atmospheric Sciences
Georgia Institute of Technology

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Education

Doctor of Philosophy in Civil and Environmental and Oceanographic Engineering (May 2024)
Massachusetts Institute of Technology (MIT) – Woods Hole Oceanographic Institution (WHOI)
Joint Program in Oceanography/Applied Ocean Science and Engineering, USA
(MIT–WHOI Joint Program)
PhD Thesis Title: Processes of Stratification Breakdown and Restratification in Antarctic Coastal Polynyas
https://hdl.handle.net/1721.1/156947

Master of Science in Civil and Environmental Engineering (April 2018)
Stanford University, USA

Bachelor of Engineering in Water Resources and Ocean Engineering (June 2016)
Zhejiang University, China

Professional Experience

Postdoctoral Fellow (October 2026 – Present), Georgia Institute of Technology, USA
Advisor: Prof. Takamitsu Ito

Visiting Scientist (July 2026 – Present), **Postdoctoral Fellow** (July 2024 – June 2026),
Johns Hopkins University, USA
Advisors: Prof. Thomas W. N. Haine (JHU) and Prof. Renske Gelderloos (TU Delft)

Graduate Research Assistant (June 2018 – April 2024), Woods Hole Oceanographic Institution, USA
PhD Advisor: Dr. Weifeng (Gordon) Zhang

Graduate Researcher (June 2017 – March 2018), Stanford University, USA
Advisor: Prof. Jenny Suckale

Undergraduate Researcher (November 2015 – June 2016), Zhejiang University, China
Advisor: Prof. Wei Li

Curriculum Vitae

Research Interests

Coastal Physical Oceanography, Numerical Ice/Ocean Modeling, Lagrangian Particle Tracking, Gravity Currents, Biological and Physical Interactions, Polynyas, Sea Ice Dynamics, Polar Marine Environments

Publications

In prep:

Xu, Y., Meng, Y., Gelderloos, R., & Haine, T. W. N. Influence of Calving-Induced Ice Mélanges on Ocean Circulation in Greenland Fjords.

Under review:

Xu, Y., Gelderloos, R., Jiang, W., & Haine, T. W. N. Influence of Subinertial Variability on Dense Overflows Across the Iceland–Faroe Ridge. Submitted to *Journal of Geophysical Research: Oceans.* A preprint is available at https://doi.org/10.22541/essoar.15003334/v2.

Published:

Xu, Y., Zhang, W. G., Maksym, T., Ji, R., Li, Y., & Walker, C. (2024). Influence of Physical Factors on Restratification of the Upper Water Column in Antarctic Coastal Polynyas. *Journal of Geophysical Research: Oceans*, 129(3), e2023JC020762. https://doi.org/10.1029/2023JC020762

Xu, Y., Zhang, W. G., Maksym, T., Ji, R., & Li, Y. (2023). Stratification Breakdown in Antarctic Coastal Polynyas. Part I: Influence of Physical Factors on the Destratification Time Scale. *Journal of Physical Oceanography*, 53(9), 2047–2067. https://doi.org/10.1175/JPO-D-22-0218.1

Xu, Y., Zhang, W. G., Maksym, T., Ji, R., & Li, Y. (2023). Stratification Breakdown in Antarctic Coastal Polynyas. Part II: Influence of an Ice Tongue and Coastline Geometry. *Journal of Physical Oceanography*, 53(9), 2069–2088. https://doi.org/10.1175/JPO-D-22-0219.1

Academic Service and Outreach

Proposal Reviewer and Panelist, U.S. National Science Foundation Office of Polar Programs (NSF OPP)

Journal Reviewer for: *Limnology and Oceanography, Journal of Geophysical Research: Oceans, Ocean Science,* and *Physical Review Research*

MIT-WHOI Joint Program Applicant Support and Knowledgebase (JP ASK), Mentor, 2022 and 2023

MIT-WHOI Joint Program, COFDL Student Meeting, Co-Organizer, 2022–2023

Gordon Research Seminar in Coastal Ocean Dynamics, Session Discussion Co-Leader, 2023

Gordon Research Seminar in Polar Marine Science, Session Discussion Co-Leader, 2023

Presentations

"Calving-induced mélange surges subduct meltwater plume intrusions in Greenland fjords", *EPS Research Day 2026*, May 1, 2026, Department of Earth and Planetary Sciences, Johns Hopkins University. (Talk)

"Influence of Calving-Induced Mélanges on Ocean Circulation in Greenland Fjords", *Ocean Sciences Meeting 2026*, February 22–27, 2026, Glasgow, Scotland. (Talk)

"Influence of sub-inertial variability on dense overflows across the Iceland-Faroe Ridge", Arctic-Subarctic Ocean Fluxes Workshop 2025, May 13-15, 2025, Instituto de Ciencias del Mar (ICM), Barcelona, Spain. (Talk)

"Influence of sub-inertial variability on dense water overflows across the Iceland-Faroe Ridge", *EPS Research Day 2025*, May 2, 2025, Department of Earth and Planetary Sciences, Johns Hopkins University. (Talk)

"Processes of Stratification Breakdown and Restratification in Antarctic Coastal Polynyas", *Atmosphere and Ocean Seminar*, September 4, 2025, Department of Earth and Planetary Sciences, Johns Hopkins University. (Talk)

"Processes of Stratification Breakdown and Restratification in Antarctic Coastal Polynyas", May 9, 2024, Ocean College, Zhejiang University. (Invited Talk)

"Processes of Stratification Breakdown and Restratification in Antarctic Coastal Polynyas", *NOAA Coastal Ocean Modeling Science Seminar*, April 23, 2024, NOAA Coastal Marine Modeling Branch. (Invited Online Talk)

"Exploring Mixing and Stratification in Antarctic Coastal Polynyas", *Parsons Laboratory Seminar Series*, March 8, 2024, Department of Civil and Environmental Engineering, Massachusetts Institute of Technology, Cambridge, MA. (Talk)

"Influence of Physical Factors on Restratification of the Upper Water Column in Antarctic Coastal Polynyas", *Ocean Sciences Meeting 2024*, February 18–23, 2024, New Orleans, LA. (Talk)

"Influence of Physical Factors on Springtime Water-Column Restratification in Antarctic Coastal Polynyas", *Gordon Research Conference on Coastal Ocean Dynamics*, June 18–23, 2023, Bryant University, Smithfield, RI. (Poster)

"Stratification Breakdown in Antarctic Coastal Polynyas", *Applied Ocean Physics and Engineering Department Seminar*, April 26, 2023, Woods Hole Oceanographic Institution, Woods Hole, MA. (Talk)

"Stratification Breakdown in Antarctic Coastal Polynyas: Influence of an Ice Tongue and Coastline Geometry", *Gordon Research Conference on Polar Marine Science*, March 5–10, 2023, Ventura, CA. (Poster)

"Influence of Physical Factors on the Stratification Breakdown and Dense Shelf Water Formation in Antarctic Coastal Polynyas", *The SCAR Open Science Conference 2022*, August 1–10, 2022. (Online Talk)

"Timescales of Wintertime Water-Column Destratification in Antarctic Coastal Polynyas", *The 10th Annual CEE Research Day*, Department of Civil and Environmental Engineering, Massachusetts Institute of Technology, April 19, 2022. (Talk)

"Timescales of Wintertime Water-Column Destratification in Antarctic Coastal Polynyas", *Ocean Sciences Meeting 2022*, February 24 – March 4, 2022. (Online Talk)

"Timescales of Stratification Breakdown in Antarctic Coastal Polynyas", *Antarctic Sea Ice and Southern Ocean Seminars*, November 18, 2020, Snow and Ice Geophysics Laboratory, University of Texas

at San Antonio, San Antonio, TX. (Online Talk)

"Numerical Modeling of Antarctic Coastal Polynya Circulation", *Gordon Research Conference on Coastal Ocean Dynamics*, June 16–21, 2019, Southern New Hampshire University, Manchester, NH. (Poster)

Workshops

The 16th International Workshop on Multi-Scale (Un)-Structured Mesh Numerical Modeling for Coastal, Shelf, and Global Ocean Dynamics (IMUM 2017), August 29 – September 1, 2017, Stanford University, Stanford, CA.

FUNWAVE-TVD Training Workshop, July 25–27, 2017, University of Delaware, Newark, DE.

Research Cruises

R/V Thomas G. Thompson, TN368, Shelfbreak Productivity Interdisciplinary Research Operation at the Pioneer Array (SPIROPA) Project Cruise, July 5–18, 2019.

SSV Corwith Cramer, MIT-WHOI Joint Program Orientation Cruise, June 29 – July 6, 2018.

Awards and Honors

Excellent Graduate of Zhejiang University and of Zhejiang Province, 2016

Outstanding Undergraduate Thesis, Zhejiang University, 2016

National Scholarship (Top 5%), Ministry of Education of the People's Republic of China, 2015

The First Prize of Excellent Undergraduate Scholarship, Zhejiang University, 2014 and 2015

References

Prof. Thomas W. N. Haine
Professor, Department of Earth and Planetary Sciences
Johns Hopkins University
Email: Thomas.Haine@jhu.edu

Dr. Weifeng (Gordon) Zhang
Senior Scientist, Applied Ocean Physics & Engineering Department
Woods Hole Oceanographic Institution
Email: wzhang@whoi.edu

Dr. Rubao Ji
Senior Scientist, Biology Department
Woods Hole Oceanographic Institution
Email: rji@whoi.edu